

Abjad AI at NADI 2026 shared task: Overcoming Data Scarcity in Arabic Speech Processing through Augmentation

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Abstract

We describe our submission to three NADI 2026 subtasks: Tunisian Arabic–French–English code-switched ASR (Task 1.3), dialectal Arabic speech translation, and 20-way Arabic dialect identification (ADI). For code-switched ASR, we finetune a 2B-parameter FastConformer encoder-decoder with synthetic Tunisian speech, cross-lingual data, waveform augmentation, and checkpoint averaging, achieving 14.12% WER and ranking first overall. For speech translation, finetuning Whisper Large-v3 yields strong in-domain results but poor generalization (BLEU 5.77 on blind test), illustrating overfitting under limited data. For ADI, a classifier on frozen pretrained ASR representations outperforms full/partial finetuning, reaching 42% accuracy across 20 dialects. We highlight synthetic augmentation, checkpoint averaging, and encoder freezing as effective low-resource strategies.

1 Introduction

Arabic dialectal variation is compounded in North Africa by pervasive code-switching with French and English, and labeled speech resources for this setting remain scarce, making data efficiency central to system design. NADI 2026 (Sullivan et al., 2026) continues the NADI shared task series (Talafha et al., 2025), evaluating dialectal Arabic speech and language processing. We describe submissions to three speech subtasks: (1) Task 1.3, code-switched ASR for Tunisian Arabic–French–English; (2) dialectal Arabic speech translation; and (3) 20-way Arabic Dialect Identification (ADI). We adapt a 2B-parameter FastConformer (Rekesh et al., 2023) ASR model and Whisper Large-v3 using strategies suited to each task: full finetuning wins for the generation tasks (ASR, translation), while a frozen encoder with a lightweight classifier wins for ADI. Our ASR system ranked **first** on Task 1.3 (WER 14.12%); our ADI system reaches 42%

accuracy; our translation system, despite strong training BLEU (44.6), generalizes poorly to blind test (BLEU 5.77), which we analyze as a negative result for low-resource speech translation.

2 Related Work

Arabic dialect identification Text-level dialect ID has been studied extensively via the NADI series (Abdul-Mageed et al., 2020; Talafha et al., 2025), while speech-based ID traces to the MGB/ADI17 challenge (Shon et al., 2020). ADI-20 (Elleuch et al., 2025a) extends ADI17 to 20 classes. Elleuch et al. (2025b) finetune Whisper-large-v3 for the coarser 8-way NADI 2025 task (79.83% accuracy); we instead freeze the encoder, which we find preferable for finer-grained 20-way classification.

Code-switched ASR Code-switching remains challenging for end-to-end ASR (Agro et al., 2026). Ben Abdallah et al. (2024) release TunSwitch, a Tunisian code-switched corpus, exploring self-supervised approaches. We instead fully finetune a large attention-based encoder–decoder model with a 48-layer FastConformer (Rekesh et al., 2023) encoder and an 8-layer Transformer decoder, using synthetic and cross-lingual data. Further details are provided in Section 3.1.

Automatic Speech Translation The IWSLT Dialectal and Low-Resource track (Anastasopoulos et al., 2022) established Tunisian-English speech translation as a low-resource benchmark; we similarly find parallel data scarcity, not model choice, is the binding constraint. Section 3.2 discusses our approach in details.

Frozen vs. Finetuned Representations SUPERB (Yang et al., 2021) showed frozen encoders competitive for tasks like language ID. We confirm this for ADI but find full finetuning wins for our generation tasks as reported in Section 4.

3 System Descriptions

3.1 (Task 1.3) Code-Switched ASR

We fully finetune Cohere Transcribe¹ model (Cohere, 2026a), a 2B-parameter encoder-decoder ASR model, on a Tunisian Arabic-French-English code-switched speech dataset provided by Task 1.3.

3.1.1 Model

An encoder-decoder model with a 48-layer Fast-Conformer (Rekesh et al., 2023) encoder and an 8-layer Transformer decoder with a 16,384-token vocabulary. It is a finetuned, Arabic-specialized derivative of Cohere Transcribe² model (Cohere, 2026b), adapted for dialectal variation and code-switching (Cohere, 2026a), which we further finetuned for Task 1.3.

3.1.2 Training

FP32 master weights with bfloat16 mixed-precision autocasting (loading directly in bfloat16 rounded small AdamW updates to zero). We use AdamW (Loshchilov and Hutter, 2019) with a cosine schedule and $\sim 3\%$ linear warmup, on 3–5 A100 80GB GPUs via PyTorch Lightning (Falcon and The PyTorch Lightning team, 2019) DDP (Table 1). All 48 encoder and 8 decoder layers remain unfrozen throughout: full finetuning consistently outperforms partial/frozen configurations for this task, a pattern also holding for speech translation. Details are given in Section 3.2.

3.1.3 Data

Training combines the official NADI 2026 Task 1.3 data with: synthetic Tunisian speech via the pipeline in Figure 1 (transcripts rewritten by Gemma 4 (Gemma Team, 2026), synthesized by OmniVoice (Zhu et al., 2026) across 25 voices, with template artifacts removed and stronger augmentation than real recordings; and 9h French (Pratap et al., 2020) + 9h English (Panayotov et al., 2015) to improve non-Arabic segment recognition. Table 1 gives per-checkpoint data composition.

3.1.4 Augmentation

We explored two augmentation strategies in separate experiments. In the first, we applied *on-line* augmentations on-the-fly during training using the audiomentations library³ (Jordal, 2022),

¹<https://huggingface.co/CohereLabs/cohere-transcribe-arabic-07-2026>

²<https://huggingface.co/CohereLabs/cohere-transcribe-03-2026>

³<https://github.com/iver56/audiomentations>

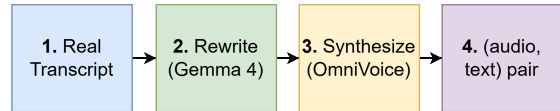


Figure 1: Synthetic data pipeline: a real transcript is rewritten by Gemma 4 for lexical variety, then synthesized by OmniVoice across 25 voices, to augment Task 1.3 training data.

with four methods: Gaussian noise, time stretching, pitch shifting, and gain perturbation, whose parameters varied per epoch. In the second, we applied *offline* augmentations, pre-generating augmented copies before training. Here we used the same four methods together with four additional ones: temporal shifting, band-pass filtering, tanh distortion, and ESC-50 (Piczak, 2015) background noise, creating five augmented copies per utterance.

3.1.5 Checkpoint Averaging

Our submission uniformly averages the weights of five checkpoints drawn from two experiments: online augmentation (CP1, CP2, CP5) and offline augmentation (CP3, CP4). These are not five independent runs: CP2 and CP5 are two checkpoints of the same online run, CP3 and CP4 are likewise two checkpoints of the same offline run (the later one taken after synthetic data was added to the training manifest), and CP1 was initialized from CP5 so the five checkpoints stem from three training runs in total (Table 1). Of these three runs, only the one producing CP1 is warm-started, from CP5’s weights; the other two start from the original pre-trained base model (Cohere, 2026a).

Weights are averaged in FP32 and cast back, following Izmailov et al. (2018). Normalization parameters and statistics (LayerNorm (Ba et al., 2016) and BatchNorm (Ioffe and Szegedy, 2015)), along with num_batches_tracked, are not averaged but taken from the best individual checkpoint among those combined.

As reported in Table 2, progressive averaging lowers WER from 16.64–16.80% for individual checkpoints to 15.97% for the 5-way average, and CER from 6.63–6.87% to 6.31%. The gain comes from combining the two experiments rather than from averaging alone: the two online checkpoints together (CP1, CP2) reach only 16.77% WER no better than the best single checkpoint (16.64%) whereas adding the first offline checkpoint (CP3) drops WER to 16.05%.

#	Training Configuration	Run	Init. Weights	LR	Aug.	Val. WER (%)
1	Real + synthetic + 9h French + 9h English	R2	From CP 5	1×10^{-5}	Online	16.64
2	Real only (early ckpt)	R1	Base	5×10^{-6}	Online	16.80
3	Real + offline aug. (early ckpt)	R3	Base	5×10^{-6}	Offline	16.80
4	+ synthetic (later ckpt)	R3	Base	5×10^{-6}	Offline	16.67
5	Real only (later ckpt)	R1	Base	5×10^{-6}	Online	16.67

Table 1: The five checkpoints combined in our final submission, with their training configuration and validation WER on the NADI 2026 Task 1.3 development set, prior to averaging.

Team	WER (%)		CER (%)	
	Raw	Norm.	Raw	Norm.
Abjad AI (ours)	14.41	14.12	5.55	5.43
Itgan	14.49	14.22	5.38	5.27
Thaka	14.53	14.26	6.34	6.26

Table 3: NADI 2026 Task 1.3 (Code-Switched ASR) official blind test leaderboard, top three teams. *Norm.* denotes Hamza normalization.

Configuration		WER	CER
Online	CP1: Real + synth. + FR/EN	16.64	6.63
	CP2: Real only	16.80	6.87
	CP5: Real only	16.67	6.68
Offl.	CP3: base	16.80	6.83
	CP4: + synth.	16.67	6.69
Avg(1,2)		16.77	6.82
Avg(1,2,3)		16.05	6.39
Avg(1,2,3,4)		16.05	6.42
Avg(1,2,3,4,5) [final]		15.97	6.31

Table 2: WER/CER (%) on the NADI 2026 Task 1.3 development set for the five individual checkpoints and their progressive averages. The final row is the 5-way average submitted on the blind test set.

3.1.6 Results

The submitted system ranked **first** on the NADI 2026 Task 1.3 blind test set. The achieved performance scores are reported in Table 3⁴. Synthetic speech, multilingual data, diverse augmentation, and checkpoint averaging together substantially improve code-switched ASR under limited-data conditions.

3.2 (Task 4) Speech Translation

We finetune Whisper Large-v3 (Radford et al., 2023) (1.55B parameters) on the shared-task data for dialectal Arabic speech translation.

3.2.1 Training Strategy

Direct finetuning on the shared-task dataset consistently outperformed (3.98 vs 5.77 Eval BLEU

Split	# Samples	Hours	BLEU
Training set	8644	10.18	–
Dev (Val) set	680	0.79	21.75
Blind test set	6780	7.78	5.77

Table 4: BLEU for our best speech translation configuration (Whisper Large-v3 finetuned on shared-task data augmented for the three weakest dialects) on the NADI 2026 training, eval and blind test sets.

score) a two-stage approach that first finetuned on an internal, out-of-domain Arabic Multi-dialects dataset of 800 hours before finetuning on the shared-task data. The domain mismatch likely outweighed the benefit of additional examples.

3.2.2 Data Augmentation

To reduce overfitting under limited data, we apply the same offline waveform augmentation (audiomentations (Jordal, 2022)) as in the previous task: Gaussian noise, time stretching, pitch shifting and temporal shifting. As before, augmented copies are pre-generated rather than applied on-the-fly. Rather than upsampling the training set, we target augmentation at the three weakest-performing dialects, generating acoustically varied copies for these subsets only. This targeted dev set contains **18/680 = 2.65%** overlap.

3.2.3 Training

We fully finetune Whisper Large-v3 with all layers unfrozen, consistent with Section 3.1, where fuller unfreezing improved performance. Our strongest configuration finetuned directly on the shared-task data augmented with the targeted three-dialect set, with no intermediate stage.

3.2.4 Results

The large train/test gap indicates substantial overfitting: targeted augmentation improves fit to limited data but lacks the diversity needed for generalization. Future work should prioritize collecting

⁴<https://www.codabench.org/competitions/17167/>

or generating genuinely diverse dialectal speech-translation data over duplicated, acoustically modified examples.

3.3 (Task 2) Arabic Dialect Identification (ADI)

We formulate ADI as 20-class utterance-level classification (MSA + 19 dialects). Our final system uses frozen acoustic representations from the same Cohere ASR encoder (Cohere, 2026a) used in Section 3.1, with a lightweight trainable classifier.

3.3.1 Model

Frame-level hidden states are mean-pooled per utterance, projected to 256 dimensions (linear + layer norm), then passed through a classification head (linear, GELU, dropout, linear) producing logits over 20 classes. The encoder remains fully frozen; only the projection and classifier are trainable. This outperformed partial or full finetuning on validation data, the opposite of the pattern for ASR and speech translation (see Section 4 for details).

3.3.2 Data and Preprocessing

We train on ADI-20 dataset; most of the official validation set is merged into training, with a small stratified subset retained for model selection. Audio is converted to mono channel, resampled to 16kHz, and RMS-normalized to discourage reliance on recording volume as a shortcut.

3.3.3 Data Augmentation and Training

Waveform augmentation (random gain, additive noise, time shifting, speed perturbation, simulated bandwidth reduction) improves robustness. We use class-balanced cross-entropy with AdamW (Loshchilov and Hutter, 2019), cosine annealing, and bfloat16 mixed precision on multiple GPUs.

3.3.4 Alternative Systems

We also explored GRU-based classification heads on Qwen3-ASR (Shi et al., 2026) and Whisper Large-v3 (Radford et al., 2023) representations. For Qwen3-ASR, a frozen encoder with augmented-only training performed best; unfreezing consistently hurt blind-test performance. For Whisper, we additionally tried intermediate training on VoxLingua107 (Valk and Alumäe, 2021), checkpoint averaging, and a zero-shot multimodal-instruction-model baseline. None outperformed the frozen Cohere-based system.

3.3.5 Results

Our system reaches **42.0%** accuracy on the blind test set, well above the 5.0% random-chance baseline for 20-way classification. Under ADI-20’s limited-data conditions, combining strong pre-trained Arabic speech representations with augmentation and a small frozen-encoder classifier consistently generalizes better than full encoder finetuning.

4 Discussion

Whether full finetuning or a frozen encoder generalizes better depends on task type, not backbone. For generation tasks – ASR and speech translation – unfreezing the full encoder/decoder consistently wins; for ADI, with less labeled data per class, a frozen encoder with a lightweight classifier wins across every backbone. Generation tasks likely benefit from adapting the acoustic-to-text mapping, while limited-data classification is more prone to overfitting when fully finetuned.

Augmentation and synthetic/cross-lingual data help, but only atop a reasonably sized authentic pool: they boosted our top-ranked ASR system, yet failed to prevent severe overfitting for speech translation’s ~2.5-hour pool – augmentation multiplies existing diversity rather than replacing it. Checkpoint averaging gave a further consistent WER/CER gain as indicated in Table 2.

5 Conclusion

We presented submissions to three NADI 2026 subtasks. Our code-switched ASR system, combining synthetic/cross-lingual data, augmentation, and checkpoint averaging, ranked first on Task 1.3 (WER 14.12%). Our ADI system shows a frozen encoder with a lightweight classifier generalizes better under limited data (42% accuracy). Our speech translation system, by contrast, illustrates overfitting under an extremely small training pool (BLEU score of 5.77).

Limitations

Our ASR system incorporates synthetic Tunisian TTS speech; despite artifact removal and stronger augmentation, some mismatch with authentic speech likely remains and its isolated contribution is unquantified. Four of five averaged checkpoints come from three training runs: two checkpoints are from one run, two from another, and the fifth was initialized from one of those checkpoints, so

the averaging benefit may partly reflect correlated checkpoints. Our speech translation system generalizes poorly (BLEU 5.77 vs. 44.58 train) and should not be taken as representative of achievable performance with more data. Our ADI backbone comparison (Cohere, Qwen3-ASR, Whisper) was not run under identical compute/tuning budgets, so rankings are suggestive; two of our three systems (ASR and ADI) share the same underlying encoder, so this comparison offers weaker evidence for a general task-type effect than fully independent backbones would. The Task 2 blind test set does not provide per-dialect labels, so we are unable to report a dialect-wise breakdown of our ADI system’s errors.

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